BIOL 222	Introduction to Food Safety and Lab	3(2, 2)
This course aims to teach the general principles of food microbiology and food safety. It covers the biology and epidemiology of foodborne microorganisms of public health significance, including bacteria, fungi, protozoa, and viruses, and food spoilage microorganisms; chemical pollution of foodstuffs; safety and preservation of food commodities; Principal methods for examination of foods; food quality control and implementation of the Hazard Analysis Critical Control Point (HACCP) scheme. The course also offers a set of experiments to cover the laboratory methods used in the microbiological analysis of foods and to identify the characteristics of the major groups of microorganisms associated with food spoilage, foodborne disease, and food fermentations. **Prerequisite**: None.		

BITM 300	Business IT Management	3(3,0)
This course provides a comprehensive introduction to information systems and their application. It explains how to use and manage information technology to revitalize business processes, conduct e-commerce, improve business decision-making, and gain a competitive advantage. **Prerequisite**: BITM 200		

CHEM 210	Principles of Chemistry	3(3,0)
This course emphasizes the fundamental principles of chemistry, including atomic structure, bonding, stoichiometry, gases, solutions, acids and bases, and solution equilibria. **Prerequisite**: None.		

CHEM 211	Environmental Chemistry and Lab	3(2,2)
This course provides an opportunity to develop an understanding of several basic environmental functions, the complicated nature of environmental systems, and the need for multidisciplinary solutions to environmental problems. Topics covered include the hydrosphere, water quantity and quality, soil and the soil ecosystem, biological resources, waste disposal, air pollution, ozone depletion, acid rain, and global warming. The course also includes a set of experiments that provide students with practical experience in various environmental analysis settings, including air and water quality. **Prerequisite**: None.		

COSC 214	Introduction to Programming	3(3,0)
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

EDUC 317	Neurological Development and Cognitive Change	3(3,0)
This course provides a unified view of neurological development, beginning with the growth of the first neural cells and extending through the emergence of complex thought and behavior. It considers brain mechanisms underlying cognitive, perceptual, social, and emotional development during infancy and childhood.		

EDUC 222	Computer Applications in Education	3(3,0)
This course focuses on general knowledge of the use of microcomputers in education, including common computer applications such as word processing, spreadsheet, database, and presentation software in teaching/learning activities, as well as evaluating the effectiveness of educational software in teaching/learning across various subject matter areas.		

GEOL 300	Human Geography: People and Culture	3(3,0)
Different cultures in different landscapes are compared to those in the Middle East. This course examines the many issues surrounding human population, perceptions of the environment, the diffusion of ideas, and the cultural aspects of resources and urban growth. **Co-requisite**: ENGL 210.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

PHYS 312	Modern Physics	3(3,0)
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and particle physics.		

Quantitative Reasoning Electives

BADM 250	Business Statistics	3(3,0)
Business Statistics introduces students to the fundamentals of applied statistics. Accordingly, students are exposed to statistics as they are directly applied in solving business problems. The course will cover random variables, sampling, probability distributions, expectation, hypothesis testing and confidence intervals, analysis of variance, correlation, and simple linear regression.		

BADM 420	Quantitative Methods for Business	3(3,0)
This course introduces students to managerial decision analysis using quantitative tools. The course will introduce students to the use and development of mathematical models to help managers make informed decisions. The focus is on the applied aspects of statistics and math. As such, the course will cover the basics of probabilistic and statistical techniques, decision analysis, linear programming, optimization, forecasting, and waiting-line theory. **Prerequisite:** BADM 250, MATH 351, or Senior Standing.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics: present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements.		

GRDS 220	Introduction to Digital Media	3(3,0)
In this course, students will develop a solid foundation in the tools and techniques of Adobe Illustrator and Adobe Photoshop. Students will learn the difference between vector-based and pixel-based programs and how to integrate both work fields to produce creative graphics. Moreover, students will be briefly introduced to Adobe InDesign, including its workspace, functions, and tools.		

GRDS 335	Production	3(3,0)
In this course, students will develop an understanding of the “process flow” of contemporary print production methods. The course covers the entire production timeline: Prepress, Press, and Postpress. In addition to printing technologies, students will learn about various types of ink and paper, binding, varnishes, finishing techniques, and the effect of the printing process on the product. Students will also learn how to handle projects and deal with clients as freelance designers or as designers working for companies. Prerequisite: GRDS 222.		

MATH 210	Discrete Mathematics	3(3,0)
Logic, propositional equivalences, predicates and quantifiers, methods of proof, proof strategy, mathematical induction, recursive definitions, strong induction, structural induction, sets and set operations, functions, growth of functions, relations and their properties, representing relations, equivalence relations, introduction to graphs, and graph terminology.		

MATH 220	Statistics for Social & Behavioral Sciences	3(3,0)
This course introduces statistical techniques for analyzing social science research data. Topics include frequency distribution, central tendency, variability, probability theory, Bayes’ rule, mathematical expectation, variance and covariance, binomial, Poisson, hypergeometric and normal distributions, sampling distributions, estimation problems, and hypothesis testing. **Prerequisite**: None.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215.		

PHIL 302	Introduction to Formal Logic	3(3,0)
This course is a standard introduction to the formal techniques of argument analysis. Formal logic was invented to mirror and evaluate mathematical reasoning; however, this course will concentrate on the relationship between formal logic and everyday reasoning and language usage in general. **Co-requisite**: ENGL 210.		

Community and Sustainability Electives

BADM 215	Personal Development and Management	1(1,0)
This course engages the student in self-reflection, self-discovery activities, and lectures. It is intended to encourage students to develop themselves and plan for the present and the future by becoming more self-aware. Topics and issues include personality types, communication styles, personal performance factors, career choices, and personal development planning.		

BADM 290	Community Engagement Experience	0(0,1)
The Community Engagement experience consists of a specific number of volunteer hours by students at community organizations under the supervision of a faculty member. Community organizations may include governmental agencies, non-governmental organizations, and civil society organizations.		

CMNS 200	Etiquette	1(1,0)
This course is designed to help students develop interpersonal and communication skills that are fundamental to success in the workplace, regardless of the industry, organization, or sector in which they are employed. Students will improve their professional style as they study topics including polite conversation, personal appearance, office politics, diplomacy, telephone, cell phone, and voicemail etiquette, meeting protocol, job interview presentations, and even international travel. Students will participate in an off-campus formal dining experience. **Co-requisite:** ENGL 210.		

CMNS 205	First Aid	1(1,0)
This course provides an overview of the principles of first aid, including preserving life, preventing further injury, and protecting the unconscious. It enhances students' knowledge of first-aid procedures and their application in real-life situations. The course also helps them develop the skills necessary to assist until medical help arrives. It raises students' awareness of different types of medical emergencies and the appropriate steps to take in both mild and life-threatening conditions. **Co-requisite:** ENGL 210.		

EDUC 315	Children's Health and Safety in the School Environment	3(3,0)
This course introduces students to childcare facilities and to environmental health concepts related to children's health and well-being. It examines current policies and practices of environmental health. It covers indoor and outdoor safety concerns, safe food handling, Universal Precautions, and other aspects of disease prevention, as well as the recognition and reporting of child abuse. It includes specific strategies to		

strengthen families to reduce the risk of child abuse and neglect. **Co-requisite** ENGL 210.

EDUC 480	**Special Topics in Education**	**3(3,0)**
This course provides students with advanced study on selected topics or emerging issues in Arab or international education. It is designed for seminar format, multimedia presentation, experiential learning activities, interactive assignments, and cooperative group work. **Prerequisite**: EDUC 201 or consent of advisor.		

ENVS 201	**Introduction to Environmental Science**	**3(3,0)**
The main goal of the Introduction to Environmental Science (ENVS201) course is to provide students (except engineering) with the scientific principles and concepts required to understand the interrelationships of the natural world, to identify and analyze environmental problems, both natural and man-made; to evaluate the relative risks associated with these problems; and to examine alternative solutions for resolving and preventing environmental problems. **Prerequisite**: None.		

GRDS 440	**Design and Social Impact**	**3(3,0)**
This course prepares students to become leading agents of social change through community engagement. It aims to promote collaboration between computer science and graphic design students to develop effective solutions to existing community issues. The course is structured around a balance of classroom theory, extensive field research, and the implementation of a real-world solution that draws on both. **Prerequisite:** Junior Standing.		

INDS 335	**Design Issues and Sustainability**	**3(3,0)**
This course investigates theoretical and philosophical constructs, as well as design issues, related to the study of interior architecture and design. It examines philosophical and practical principles of sustainable design by exploring environmental issues, sustainable materials and methods, and professional practice. Students will develop an awareness of the environmental implications of design decisions. They will gain a foundation for evaluating materials, processes, and practices in accordance with sustainable and environmentally responsible design principles.		

PSYC 314	**Psychology for Sustainability**	**3(3,0)**
This course will take a holistic approach to the topic, drawing on most major areas of psychology to explore environmental sustainability from a psychological perspective. We will use psychological principles, theories, and methods to examine the reciprocal relationship between human beings and the natural world, discuss the complex nature of environmental problems, and review important psychological and social factors that underlie sustainable and non-sustainable behaviors. **Co-requisite**: ENGL 210.		